

A geographical information system based multi-criteria decision-making approach for location analysis and evaluation of urban photovoltaic charging station: A case study in Beijing

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ABSTRACT

The integration of photovoltaic (PV) power generation system and electric vehicle (EV) charging station can effectively promote the local consumption of renewable energy and reduce the indirect carbon emissions of EV. This paper aims to provide a practical model for location decision of PV charging station (PVCS) which combines geographic information system (GIS) with multi-criteria decision making (MCDM) methods. To verify the feasibility and practicality, an empirical study was conducted in Beijing. First, seven suitable areas were selected preliminarily by suitability analysis of GIS. This stage focused on traffic flow and road distribution. Second, MCDM methods were used to further evaluation. At this stage, a comprehensive evaluation index system of natural, economic, technical and social criteria was established. Interval number, triangular fuzzy number and hesitant fuzzy linguistic term set were utilized to collect and describe evaluation information. Then, the subjective and objective weights of the criteria were calculated by best-worst method and mixed information entropy method, respectively. Finally, the TODIM (an acronym in Portuguese of interactive and MCDM) method was used to rank the alternatives. The ranking results illustrated that alternative sites A3 and A7 are outstanding. Moreover, dual sensitivity analysis and comparative analysis proved the stability and reliability of the model. Scenario analysis expanded the application scope of TODIM method, which also showed that it is necessary to express decision preferences by setting different recession coefficients. This study can provide support for the layout of PVCS in urban, and enrich the application fields of GIS and MCDM methods.

1. Introduction

The transportation sector is an important source of carbon dioxide production [1]. Under the dual pressures of energy and environment, electric vehicle (EV) came into being and received extremely close attention from the market and researchers. The “EV Charging Infrastructure Development Guide (2015–2020)” issued by the National Development and Reform Commission shows that the goal of China's charging infrastructure development is to build 12,000 centralized charging stations and 4.8 million charging piles by 2020 to meet the needs of 5 million EVs in the country [2]. However, the sustainable development of EV still faces various obstacles and challenges [3]. First, EV has shorter driving mileage and lacks a complete charging infrastructure [4]. Driving mileage needs to be solved by further scientific technology. Under the circumstance of limited mileage, more scientific and reasonable planning and construction of charging infrastructure to

improve its utilization efficiency and convenience is the key to promote the development of EV [5]. Second, coal is still the main primary energy source in the power generation side of China's power system. EVs are directly connected to the grid through the charging infrastructure, and the actual indirect carbon emissions are not significantly superior to conventional fuel vehicles [6].

Faced with the above problems, it is an effective way to combine charging infrastructure with renewable energy power generation system to achieve true low carbon and sustainable development [3,7]. The development of renewable energy utilization technology promotes the integration of photovoltaic (PV) power generation system and EV charging station (EVCS) [8]. The EVCS has stable power demand, and PV power generation equipment can provide power. These characteristics are the key to the combination and complementarity of them. In addition, the planning idea of PV power generation project is “nearly using the surplus to power grid”. Correspondingly, PV charging stations

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(PVCSSs) can promote the local consumption of PV power, reduce the dependence of charging stations on the grid, and directly reduce carbon emissions [3]. Therefore, the layout of PVCSS would contribute to the sustainable development of EVs and society. PVCSS has attracted the close attention of researchers. Tulpule et al. [9] analyzed the economic and environmental impact of PVCSS by comparing the optimal charging scenario with uncontrolled charging scenario. Han et al. [10] presented an economic evaluation method for PVCSS using retired EV batteries. Chaudhari et al. [11] proposed a hybrid optimization algorithm for energy storage of PVCSS. Mouli and Bauer [12] designed a PVCSS based on the idea of maximizing revenue. Farhadi [13] designed a novel grid-connected PVCSS, which uses maximum power point tracking technology. Guo and Zhao [14] and Wu et al. [15] used multi-criteria decision-making methods to study the location of EVCS. But the research object is ordinary charging station. It can be seen these studies are mainly focusing on optimizing allocation and charging strategy and there are few studies on location decision of PVCSS.

However, the location decision of PVCSS is the prerequisite to solve the above problems. The specific reasons can be summarized as follows: Firstly, a scientific and reasonable site can reduce the cost of construction and operation, thereby improving the economic benefits of the investors. Secondly, only by choosing the suitable site can more users be served, thereby improving public recognition and satisfaction. Thirdly, the scientifically chosen location can provide an important guarantee for future expansion and sustainable development. Therefore, how to lay out scientifically and reasonably is a subject worthy of deep study. In the end, the aim of this paper is to provide investors and decision makers with a practical and reasonable approach for location decision of urban PVCSS.

At present, commercial charging stations can be divided into ordinary charging stations and fast charging stations: ordinary charging stations are mainly distributed in residential areas, public parking lots and commercial buildings, which are closely integrated with parking spaces; fast charging stations are similar to gas stations, which are attached to main roads according to traffic flow layout. The PVCSS studied in this paper belongs to the fast charging station, which mainly serves the pure EV. The layout of the PVCSS should focus on the urban road layout and traffic flow. Therefore, this paper will use Geographic Information System (GIS) technology to analyze the urban road distribution and traffic flow, to provide a preliminary basis for site selection. Then, the MCDM methods are used to further optimize the sites.

The originality of this paper can be summarized as follows: 1) Based on the latest geographic information data (road distribution, traffic flow, etc.), GIS technology is used to initially screen out the sites suitable for PVCSS in Beijing. 2) Through a deep study of related literature and industry reports, supplemented by expert consultation, the criteria system for further evaluation of the alternative sites is established. The criteria system covers four groups: natural, economic, technical and social factors. 3) In order to make the evaluation results more accurate, this paper uses the corresponding information description tool to collect and express the index attribute values: the interval number is used to describe the quantitative economic indicators to reflect the influence of price and rate fluctuations; the hesitant fuzzy linguistic term set (HFLTS) is used to collect evaluation information of qualitative indicators. Then, the HFLTS can be converted into triangular fuzzy numbers for quantitative calculation and analysis. 4) In this paper, TODIM method based on prospect theory is used to rank alternative sites selected by GIS technology. This method can represent the decision preference of decision makers for risk and loss avoidance. In addition, this paper extends the using environment of this method to the environment of mixed index attribute values, and sets different recession coefficients for different indicators to reflect different risk preferences of decision makers. The result of scenario analysis shows that this application change can extend the using scenario of TODIM method to a state closer to the actual decision-making. 5) Based on the above methods and theories, this paper constructs a practical analysis and

decision-making framework for the location decision of urban PVCSS.

To this end, the rest of the paper is organized as follows: In view of the main points of this study, a literature review is carried out in Section 2. Section 3 introduces the geographic information data used in the stage of GIS suitability analysis and the criteria system for further evaluation of alternative sites. Section 4 introduces the main methods and related theories used in this paper. In Section 5, based on the proposed location decision framework, an empirical study is carried out to verify the overall model and framework of this paper. Further analysis and discussion are carried out in Section 6. Dual sensitivity analysis and comparative analysis are used to test the stability and applicability of the model. Through scenario analysis, this paper studies the changes of final results after different criteria adopting different recession coefficients. Conclusions are provided in Section 7.

2. Literature review

2.1. Location decision criteria for PVCSS

Through the analysis of a large number of literatures, it can be found that the object of their topic is basically the conventional EVCS. Scholars have conducted extensive research on the location of EVCSs by considering various influencing factors and different decision-making perspectives. These research results provide important support for the location decision of PVCSS studied in this paper.

Social, economic and environmental factors are the key considerations for scholars to plan the location of EVCS. On this basis, Lin et al. [16] established the weight coefficient of charging station. Guo and Zhao [14] evaluated the alternative charging station sites from three aspects of economy, society and environment based on the sustainable perspective. Wu et al. [15] optimized the appropriate site for EVCS from the perspectives of economy, society, environment and planning. Zhao and Li [17] screened out the optimal initial index of charging station location from the perspective of extended sustainability theory, and further determined the key index by Delphi method. Finally, they established an index system consisting of economic, social, environmental and technological aspects. Qing et al. [18] considered the constraints of land price, construction cost, operation and maintenance cost, traffic flow and service scope, and established a charging facility planning model. Sadeghi-Barzani et al. [19] proposed a hybrid integer non-linear optimization method for the optimal configuration of fast charging stations. The optimization method considers the development cost, energy consumption loss, power grid loss, the location of charging station and urban road. Besides, they pointed out that the reliability of power grid could affect the position of charging station. In addition to the above factors, Yan and Ma [20] also considered the market share of the EV industry, the safety of charging facilities and the technical maturity of the EV to provide a basis for the location of the EVCS. Davidov and Pantoš [21] believed that charging reliability and service quality are the key to the layout of charging facilities for EVs. YAĞCITEKİN et al. [22] focused on the convenience and scalability of the charging station.

Through the literature review and analysis above, it can be seen that the location decision of PVCSS should consider the economic, social, technological, traffic flow, and geographical distribution factors. Therefore, the location decision of PV power plants should be a MCDM problem. In addition, due to the natural particularity of the PVCSS, the solar energy resources and temperature of the alternative site should also be considered. The location decision factors and criteria considered in this study are summarized in Table 1. Besides, Section 3 will elaborate on the specific meaning of each factor and criterion.

2.2. Multi-criteria decision making methods for location decision

The main multi-criteria decision-making (MCDM) methods used in location decision can be divided into two categories: Weight

Table 1

The location decision factors and criteria considered in this study.

Consideration mentioned in the literature	Factors and criteria constructed in this study	Analytical methods
Urban Road Distribution	Main Road distribution	GIS technology
Traffic flow	Interest points, Land use types	
Planning layout	Existing charging station layout	Multi-criteria decision making
Security	Waterway distribution, Gas station distribution	
Scalability	Possibility of capacity expansion in the future	
Economic performance	Construction cost, Payback period	
Technical factors	Impact on the power grid	
Social factors	Government support, Public recognition	
Natural environment	Direct normal irradiation, Annual average temperature	

determination methods and ranking methods. The former mainly include entropy method, AHP (Analytic hierarchy process), ANP (Analytic network process), DEMATEL (Decision Making Trial and Evaluation Laboratory), BWM (Best-worst method). The latter mainly include TOPSIS (Technique for Order Preference by Similarity to an Ideal Solution), VIKOR (Vlsekriterijumska Optimizacija I Kompromisno Resenje), PROMETHEE (Preference Ranking Organization Method for Enrichment Evaluations), ELECTRE (Elimination et Choix Traduisant la Realite) and TODIM (an acronym in Portuguese of interactive and MCDM). The main characteristics of these methods are summarized in Appendix 1.

In the determination of weights, both subjective and objective weights should be considered. Therefore, the combined weighting method is more suitable for determining the evaluation criteria of this location decision problem. BWM can greatly reduce the number of comparisons and the probability of comparison errors [23,24]. It has been widely used in various research fields: determining the weight of green innovation obstacles [25], the weight of speaker prototype design selection criteria [26], the weight of biomass conversion technology selection criteria [27], olive harvester selection criteria [28], the weight of passenger experience criteria [29]. Besides, there is not obvious correlation among the evaluation criteria constructed in this paper, and the number of indicators is relatively moderate. Considering that the objective weight is easy to calculate, this study chooses BWM and entropy method to determine the subjective weight and objective weight respectively, and uses linear weighting method to fuse the calculation results.

In the actual location decision process, due to the influence of cognitive ability, emotional and psychological factors, decision-makers often have limited rational psychological characteristics, and there is a gap in their perception of the benefits and losses of each criterion. Therefore, compared with other ranking methods in Appendix 1, the prospect theory represented by TODIM is more suitable for location decision. The validity of TODIM method lies in its solid mathematical foundation and simple application, which can obtain decision-making results in line with the preferences of decision makers. Therefore, TODIM method has been successfully applied to the field of location decision, such as airport site selection [30], landfill site selection [31], distributed PV project [32]. At the same time, it is necessary to extend the traditional TODIM method to the decision-making environment of mixed data type, so as to better solve the actual location decision problem of urban PVCS.

2.3. Combined application of GIS and multi-criteria decision making methods

As a mature platform of geospatial data processing and analysis, GIS has received attention in various fields. Location analysis is a specific application of GIS, which has been widely used in many researches. This paper will study the combination of GIS and MCDM.

Tahri et al. [33] and Uyan [34] used GIS technology and AHP method to evaluate the location of solar farms. The combination of TOPSIS, AHP and GIS has been applied to the location decision of solar

farms [35] and wind farms [36]. Vasileiou et al. [37] also combined the AHP method with GIS to study the location of Greek hybrid offshore wind and wave energy systems. Hohn et al. [38] analyzed the potential biomasses and sites for biogas plants in southern Finland by using regional assessment method combined with GIS. Atici et al. [39] used GIS based MCDM method to study the location decision of wind farms. Gigović et al. [40] combined DEMATEL method with GIS to analyze Serbia's suitable location for tourism development. Giamalaki and Tsoutsos [41] used GIS and AHP methods to identify and determine the site suitable for installing solar power. The first step is to use GIS to identify suitable areas, and then use AHP to sort the alternative areas. Kamdar et al. [42] also used AHP method combined with GIS to study the suitable site of Municipal solid waste landfill. Ramya and Devadas [43] used MCDM methods and GIS to identify areas suitable for industrial development. The combinations are also used by Konstantinos et al. [44] to select suitable sites for wind farm layout. There are many cases and studies on the application of GIS and MCDM. Most of the above mentioned are related research in the energy field (solar and wind farms). Through the results of existing research, it can be concluded that their combined application is mature and effective.

GIS provides unique capabilities for automated geographic analysis and provides efficient data processing and description. MCDM is effective in dealing with the lack of valid data problems and merging subjective judgments into the decision-making process, and ultimately provides alternatives ranking result based on multi-criteria. Therefore, this paper will use the MCDM methods based on GIS to study the location decision of urban PVCS. Different from the above researches, this paper uses the TODIM method to perform the second stage of ranking. It is worth pointing out that in the multi-criteria evaluation stage, this paper sets up various scenarios to represent the different risk preferences of decision makers.

2.4. Findings of the literature review and contribution of this paper

Through the literature review above, it can be found that some issues have not been solved well by existing studies:

- (1) There is currently no research on location decision for urban PVCS. The object of this research field is basically the general EVCS, without considering the combination with renewable energy power generation.
- (2) In the existing decision researches of EVCS location, the alternative sites are often fictitious, lacking reliable sources and scientific access. This reduces the applicability and completeness of the whole study.
- (3) The accuracy of the entire decision model needs to be improved by improving the evaluation information acquisition process.
- (4) There is a need to improve the use environment of the MCDM method to match the actual decision scenario.

In order to solve the above issues, this paper makes the following improvements:

- (1) The research object of this paper is PVCS. Different from the previous location decision of EVCS, the alternative sites of this study are obtained by using GIS through suitability analysis, which can fully consider the urban geographic information. Then, the MCDM methods are used to rank the alternative sites.
- (2) This paper uses interval numbers to describe quantitative economic indicators to reflect the impact of price and rate fluctuations. The hesitant fuzzy linguistic terminology set is used to collect qualitative evaluation information.
- (3) In view of the mixed type of evaluation information in this study, the entropy and TODIM method are extended to the mixed attribute value environment. In addition, the extended usage of TODIM method will be carried out through scenario analysis.

3. Location analysis and evaluation criteria for photovoltaic charging station

The determination of analysis and evaluation criteria is the basic work for selecting the suitable site of PVCS. One of the key contents of this paper is to construct two-stage criteria to prepare for the follow-up analysis work. Through consulting a large number of literature and industry reports (Table 1), and discussing with experts in energy development and EV related fields, these criteria were determined. In the first stage, based on a large number of detailed GIS data, this paper uses ArcGIS software to carry out suitability analysis, so as to preliminarily screen out the site with high suitability. In this stage, key geographic information such as roads, traffic, rivers, land use and interest points are mainly used. Based on the suitable areas analyzed in the first stage, this paper establishes four groups of evaluation criteria including natural factors G1, economic factors G2, technical factors G3 and social factors G4 in the second stage to further analyze the specific site suitable for the construction of PVCS. The specific GIS data and evaluation criteria used in the two stages are described below.

3.1. Data used for suitability analysis in GIS

- Land use types [33–36,39]. This kind of GIS data can reflect the land type of the area to be built. It can be divided into park, forest, farm, commercial land, industrial land, military zone, residential area, cemetery, orchard, shrub and other types. It is worth pointing out that alternative stations should be located in areas with human, building and traffic flows, so the barren and sparsely populated areas are excluded from this kind of data. The land use data used in this paper contains 18 classifications, which can serve the suitability analysis well. For the PVCS, the suitability of each type of land is different. 10-point scale is set in this paper. The closer the value is to 10, the better the suitability is. On the contrary, the closer the value is to 1, the worse the suitability of this kind of land is. For example, the suitability of military zones [39] and nature reserves is poor, and that of shrubs is better.
- Road distribution [33,34,39]. The research object of this paper is PVCS, whose layout planning is similar to that of gas station to a certain extent, and is located near the main road according to traffic flow. Therefore, the suitability under this standard can be reflected by the distance between the selected area and the main road.
- Waterway distribution [35,36,39]. This kind of data reflects in detail the distribution of rivers and waterways in the whole area to be selected. Considering the safety, the choice of charging station should be as far away from the water source as possible. Therefore, the suitability under this index can be reflected by the distance between the selected area and the waterway.
- Gas station distribution. This kind of data reflects in detail the distribution of gas stations in the whole area to be selected. First of all, considering the safety, the charging station of EV should be as far away from the gas station as possible. In addition, in order to avoid traffic congestion and waste of road resources, EV supply points

should be separated from fuel vehicle supply points. Therefore, the suitability under this index can be reflected by the distance between the selected area and the existing gas station.

- Interest points [35,36,39,40]. The point of interest is a landmark and attraction on the electronic map, which is used to indicate the government departments represented by the place, the commercial institutions of various industries, tourist attractions, historic sites, transportation facilities and other premises (such as park, playground, school, hotel). The more interest points in a region, the greater the traffic demand, the greater the charging demand of electric vehicles. Decision makers should give priority to the layout of PVCS in such regions.

Appendix 2 visually shows the geographic information map generated by the relevant GIS data used in this study. In the fifth chapter, based on GIS technology, suitability analysis will be used to preliminarily determine the suitable site for the construction of PVCS.

3.2. Further evaluation criteria for the location of PVCS

3.2.1. Natural factors G1

- Direct normal irradiation [32,45–48]. Solar energy resource is an important factor that affects the location of PVCS. The level of solar energy resource would directly affect the power generation of PVCS, and then influence the economic benefit of the project. The research area of this paper is the whole city of Beijing. Overall, the difference of solar energy resources in the whole city is not obvious. Therefore, this criterion was not used in the GIS stage. It is more intuitive to characterize the difference by collecting the specific values of the direct normal irradiation from the selected area.
- Annual average temperature [32,49–51]. Radiation and temperature are the main factors affecting the efficiency of PV power generation. When the temperature of solar cell module is high, the open circuit voltage decreases dramatically with the increase of temperature. In extreme cases, the charging point is seriously offset, which may cause the system to be undercharged and damaged. The output power of solar cells will also decrease dramatically with the increase of temperature, which makes the solar cell module unable to give full play to its maximum performance. In addition, the batteries used in the project would also be affected by temperature.

3.2.2. Economic factors G2

- Construction cost [14,15,52,53]. The PVCS can be divided into five sub-modules: PV power generation system, distribution system, charging system, battery dispatching system and charging station monitoring system. Correspondingly, the construction cost of PVCS is mainly composed of three parts: land acquisition fee, infrastructure cost, distribution facility cost. As a functional building with new concepts, the design and construction of PVCS need to consume a lot of funds. It is worth noting that the charging station will be built in the urban area with higher land prices, which is an important factor that investors have to consider. Therefore, the construction cost is a very important evaluation criterion.
- Payback period [14,15,32,54]. The payback period of PVCS investment refers to the time when the net income of the project offsets the total investment. The length of the payback period represents the risk in a certain degree. Therefore, it is an important index that needs the consideration of decision makers. And the calculation of investment payback period of PVCS project involves the construction cost, operation and maintenance cost, PV subsidy policy, charging standards, electricity need prediction for electric vehicles, etc. This paper would focus on the above aspects to study the payback period of PVCS.

3.2.3. Technical factors G3

- Impact on the power grid [32,55,56]. When the PV power cannot meet the charging demand of EV, the distribution network will be used to charge EV together. In general, in order to prevent the power supply line from being overloaded, special high-voltage transmission network and step-down transformer are needed before the power grid is connected to the fast charging station of EV. High-voltage power is directly transmitted to the EVs through the transformer and the alternating current-direct current converter. Because there is no power buffer between them, the high-power and short-time charging characteristics of fast charging stations will affect the stability of power grid transmission, interfere with other users and even cause power accidents. Therefore, it is necessary to consider the impact of the charging station on the grid.
- Possibility of capacity expansion in the future [15,57,58]. With the improvement of people's environmental awareness, EV has entered the public's vision more widely. It can be imagined that in the near future, more and more people will drive EV. Therefore, it is necessary to incorporate the possibility of capacity expansion into the criteria system of PVCS construction. The expansion of the capacity of charging stations needs to consider the number of charging users in the future, land resources nearby, government policies, and the upgrade of distribution network.

3.2.4. Social factors G4

- Government support [32,59,60]. The PVCS project has a large initial investment cost and a long payback period, which is highly vulnerable to the influence of government policies. Specifically speaking, the approval of construction land, the upgrading and transformation of distribution network, the implementation of subsidy policy and the traffic planning near the charging station all need the support of the government. Therefore, government support is one of the indicators which is worth considering in this study.
- Public recognition [61,62]. The PVCS would inevitably affect the local environment, transportation, power grid and so on in the early construction process, and also bring certain light pollution after the project is completed. But in the long run, the project will bring enormous economic and social benefits. It will not only benefit the environmental protection, but also promote the change of consumption concept of nearby residents. People may choose to purchase EV instead of traditional car because of the convenience of charging stations. The public could comprehensively analyze the pros and cons of the project and feedback their own attitudes toward the PVCS. Public awareness and support will affect the development of similar projects and the future development speed of EV.

According to the above specific analysis and elaboration, the further evaluation index system constructed in this paper are shown in Fig. 1.

4. Methodology

4.1. Geographic information system (GIS)

Geographic Information System, also referred to as GIS, is a comprehensive subject. Combining geography, cartography, remote sensing, surveying and mapping, and computer science, GIS is widely used in cartography, navigation, logistics, urban planning and other fields [63]. GIS is composed of spatial data, system hardware, system software and users. According to its functions, the software of GIS system can be divided into five parts: data input, data management, spatial analysis, data output and application module. ArcGIS is the world's leading GIS construction and application platform, but also the highest market share of the Global GIS software. This study uses the ArcGIS 10.2 Desktop software launched by ERSI in 2013 to achieve suitability

analysis. Thematic layer is the basic framework of data analysis in GIS. GIS usually adopt the principle of classifying geographical objects by subject and layering them by category. The collection of the same elements is a thematic layer. Different themes are formed by overlapping different layers.

GIS technology has been applied in many fields, such as surveying and mapping science, resource investigation, environmental assessment, transportation, water conservancy and electricity, and plays an increasingly important role [35]. GIS can well match decision-making technology to provide decision-makers with more scientific and intuitive data analysis and data management functions. Geographic information data can be stored, managed, analyzed and visualized by GIS, so as to provide the initial suitable region or object for the evaluation of MCDM stage. GIS has been more and more used for spatial suitability analysis and modeling, which is an important means of location modeling using spatial data [64]. Therefore, this study uses GIS to make a preliminary suitability analysis of the area where the PVCS is laid out, so as to serve for further determining the optimal site.

This study mainly uses the following key steps to analyze the suitability of the location of PVCS. Firstly, distance analysis is carried out to calculate the Euclidean Distance of criteria elements corresponding to road layer, water layer and gas station layer based on the acquired geographic information data (Appendix 2). The second step is to reclassify land use types, road distribution, waterway distribution and gas station distribution. In this part, the 10-point scoring method was used. For example, the closer the distance to the main road, the better the suitability, and the score given by the reclassification is closer to 10 points; the distribution of gas stations and waterways is reversed. The available land could be assigned different scores depending on whether it is suitable for building a PVCS. The third step is suitability analysis. After reclassification, each data set is unified into the same hierarchy, and the attributes considered appropriate in each data set are given higher values. Four factors are given different weights, and then data sets are merged to form intuitive maps for suitability analysis. The weighted formula for the final suitability data set is as follows:

$$\text{Suitability} = \alpha_1 \text{Reclassroad} + \alpha_2 \text{Reclassland} + \alpha_3 \text{Reclasswater} + \alpha_4 \text{Reclassgas} \quad (1)$$

where α_1 , α_2 , α_3 and α_4 represent the weight of these four factors and $\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 = 1$.

Then, regions with suitability greater than 6 are selected for further analysis. Besides, compare the number of interest points in these regions, and consider the distribution of existing charging stations. The area with a large number of interest points indicates that the charging demand of EVs is greater, and the layout of PVCS should be given priority. Finally, through the above operation and analysis, the alternative sites to be selected suitable for building the PVCS can be obtained.

4.2. Collection and description of evaluation information

4.2.1. Interval number

Definition 1.. If $\tilde{x} = [x^l, x^u] = \{x \mid x^l \leq x \leq x^u, x, x^l, x^u \in R\}$, then $\tilde{x}$ is called interval number (IN). [65,66]

where, x^u and x^l are the upper and lower limits supported by the interval number $\tilde{x}$. Besides, if $\tilde{x} = [x^l, x^u]$ also satisfies $0 < x^l \leq x^u < 1$, then $\tilde{x}$ is called a normative IN.

Distance calculation: If interval numbers $\tilde{a} = [a^l, a^u]$ and $\tilde{b} = [b^l, b^u]$ are known, the distance $D(\tilde{a}, \tilde{b})$ between them can be obtained by the following formula:

$$D(\tilde{a}, \tilde{b}) = \sqrt{\frac{(a^l - b^l)^2 + (a^u - b^u)^2}{2}} \quad (2)$$

Normalization of Interval Numbers: $D(\tilde{a}, \tilde{b})$ is known as the interval number in the decision matrix. Among them, i represents the i^{th} decision

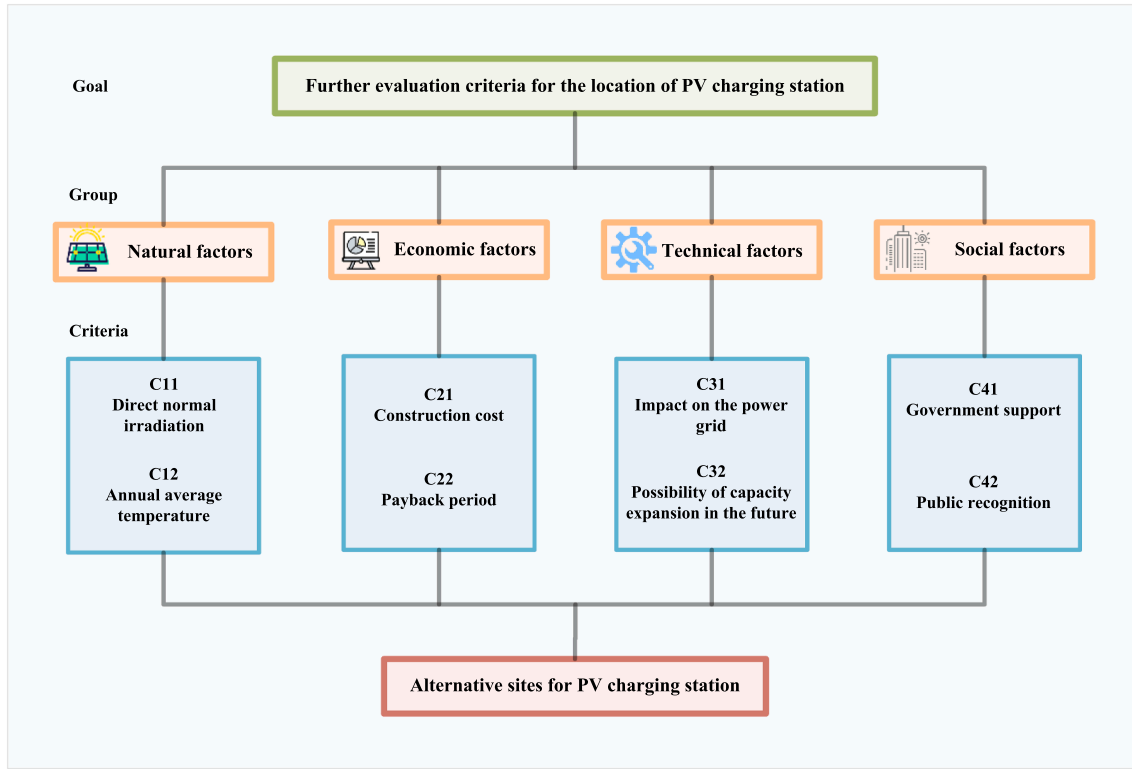


Fig. 1. Further evaluation criteria for the location of PV charging station.

attribute and j represents the j^{th} alternative. The normalization method is as follows:

$$\text{Profit index: } \left[\frac{a_{ij}^l}{\max a_{ij}^u}, \frac{a_{ij}^u}{\max a_{ij}^u} \right] \quad (3)$$

$$\text{Cost index: } \left[\frac{\min a_{ij}^l}{a_{ij}^u}, \frac{\min a_{ij}^l}{a_{ij}^l} \right] \quad (4)$$

4.2.2. Triangular fuzzy number

Definition 2.. [67] If $\tilde{x} = [x^l, x^m, x^u] = \{x \mid 0 < x^l \leq x^m \leq x^u, x, x^l, x^m, x^u \in R\}$, then $\tilde{x}$ is called Triangular fuzzy number (TFN).

where, x^u and x^l are the upper and lower limits supported by the triangular fuzzy number $\tilde{x}$. x^m is the median supported by the $\tilde{x}$. x^m is also called information preference value, usually the number with the largest discipline selected in the region. Besides, if $\tilde{x}$ also satisfies $0 < x^l \leq x^m \leq x^u < 1$, then $\tilde{x}$ is called a normative TFN.

If $x^l = x^m$ or $x^m = x^u$, then $\tilde{x} = [x^l, x^m, x^u]$ degenerates into an interval number $\tilde{x} = [x^m, x^u]$ or $\tilde{x} = [x^l, x^m]$.

Distance calculation: If two TFNs $\tilde{a} = (a^l, a^m, a^u)$ and $\tilde{b} = (b^l, b^m, b^u)$ are known, the distance $D(\tilde{a}, \tilde{b})$ between them can be obtained by the following formula:

$$D(\tilde{a}, \tilde{b}) = \sqrt{\frac{(a^l - b^l)^2 + (a^m - b^m)^2 + (a^u - b^u)^2}{3}} \quad (5)$$

4.2.3. Hesitant fuzzy linguistic term sets with credibility

In practical decision-making, due to the incompleteness and uncertainty of information, decision makers often hesitate among several possible linguistic terms when evaluating the attributes of the selected objects. To solve this problem, Rodriguez et al. proposed the concept of hesitant fuzzy linguistic term set (HFLTS). It increases the flexibility and application scope of evaluation linguistic term, and can collect and

obtain evaluation information of qualitative attributes, especially risk indicators. The relevant definitions to be used are as follows:

Definition 3.. Linguistic term set $S = \{s_i, s_{i+1}, s_{i+2}, \dots, s_n\}$ is a finite ordered set, and the number of linguistic terms contained in S is odd [68]. The corresponding language terminology set S of HFLTS used in this paper is defined as follows:

$$S = \left\{ \begin{array}{l} s_0: \text{very Low (VL)}, s_1: \text{Low (L)}, s_2: \text{Relatively Low (RL)}, \\ s_3: \text{Medium (M)}, \\ s_4: \text{Relatively High (RH)}, s_5: \text{High (H)}, s_6: \text{Very High (VH)} \end{array} \right\} \quad (6)$$

Definition 4.. [69] Given the linguistic terminology set S , if H_S is a set of finite sequential linguistic terms in the linguistic terminology set S , then H_S is called a HFLTS on S . The mathematical form is:

$$H_S = \{ \langle x, h(x) \rangle \mid x \in X \} \quad (7)$$

where function $h(x)$ denotes the possible membership of $x \in X$ mapped to set H_S , and $h(x)$ is a series of possible values in linguistic term set S .

Definition 5.. [69] E_{GH} represents a transformation relationship, which can convert the evaluation description (ED) given by experts according to the linguistic terminology set S into the corresponding HFLTS H_S . The logical relationship is as follows:

$$E_{GH}: ED \rightarrow H_S$$

The specific rules are as follows:

$$E_{GH}(s_i) = \{s_i \mid s_i \in S\} = \{s_i\} \quad (8)$$

$$E_{GH}(\text{between } s_i \text{ and } s_j) = \{s_k \mid s_k \in S, s_i \leq s_k \leq s_j\} = \{s_i, s_{i+1}, \dots, s_j\} \quad (9)$$

$$E_{GH}(\text{below } s_i) = \{s_k \mid s_k \in S, s_k \leq s_i\} = \{s_0, s_1, \dots, s_i\} \quad (10)$$

$$E_{GH}(\text{above } s_i) = \{s_k \mid s_k \in S, s_k \geq s_i\} = \{s_i, s_{i+1}, \dots, s_6\} \quad (11)$$

When respondents evaluate the target criteria according to a given linguistic term set, they tend to prefer a linguistic term or some

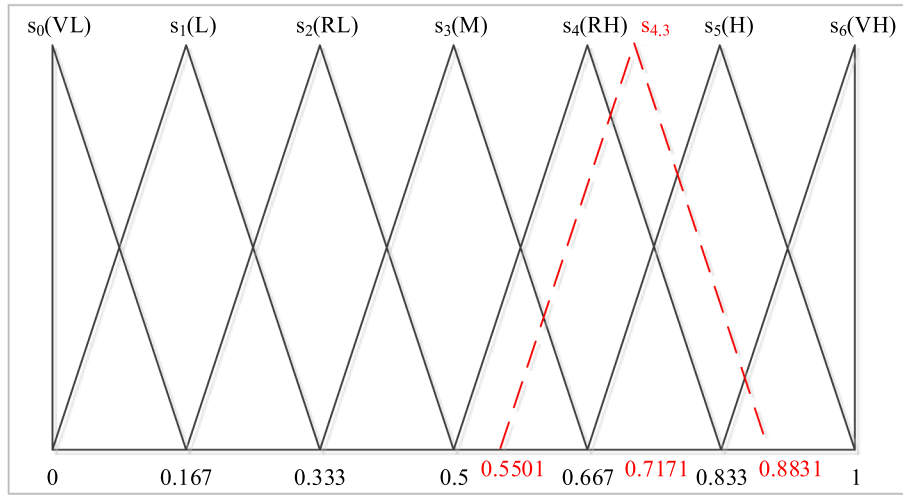


Fig. 2. The corresponding relationship between HFLTS and TFNs.

linguistic terms. Therefore, this paper uses the HFLTS with credibility [45,70] to collect the evaluation information of qualitative criteria.

Example 1.. Respondents consider that the public acceptance C42 of the alternative site A1 is between medium and high level. Use the transformation rules introduced in **Definition 6**. The HFLTS with credibility for this qualitative criterion can be obtained as follows:

$$H_S(C_{ij}) = \left\{ \frac{0.3}{M}, \frac{0.6}{RH}, \frac{0.1}{H} \right\}$$

The credibility of the linguistic value for evaluating the public acceptance C42 are 0.3, 0.6 and 0.1, which correspond to the medium, relatively high and high respectively.

After obtaining HFLTS with credibility, in order to facilitate further quantitative analysis and calculation, this paper further converts it into TFNs. The corresponding relationship between them is shown in Fig. 2 [45,70].

Example 2.. Taking $s_{4,3}$ as an example:

$$\begin{aligned} s_{4,3}: & (0.5501, 0.7171, 0.8831) \\ &= (0.5 + (1.3 - 1) \times 0.167, 0.667 + (1.3 - 1) \times 0.167, \\ &0.833 + (1.3 - 1) \times 0.167) \end{aligned}$$

According to the symmetric operation rules of subscripts, the correspondence between linguistic term set S and TFNs is obtained. The results are as follows:

$$S = \left\{ \begin{array}{l} s_0: (0, 0, 0.167), s_1: (0, 0.167, 0.333), \\ s_2: (0.167, 0.333, 0.5), s_3: (0.333, 0.5, 0.667), \\ s_4: (0.5, 0.667, 0.833), s_5: (0.667, 0.833, 1), s_6: (0.833, 1, 1) \end{array} \right\}$$

Through the above theoretical analysis, HFLTS with credibility can

be transformed into TFNs, and then can be calculated quantitatively according to the operation rules of TFNs [71]. The further processing of Example 1 is as follows:

$$\begin{aligned} H_S(C_{ij}) &= \left\{ \frac{0.3}{M}, \frac{0.6}{RH}, \frac{0.1}{H} \right\} \\ &= (0.3s_3, 0.6s_4, 0.1s_5) \\ &= (0.3 * (0.333, 0.5, 0.667), 0.6 * (0.5, 0.667, 0.833), 0.1 \\ &\quad * (0.667, 0.833, 1)) \\ &= (0.4666, 0.6335, 0.7999) \end{aligned}$$

4.3. Multi-criteria decision making methods

4.3.1. Best-worst method (BWM)

The BWM is a method of determining subjective weights. Generally, the MCDM problem needs to compare the indicators in pairs, and then obtain the evaluation matrix of the indicators. In the process of consistency test, if the consistency results of evaluation matrix cannot meet the requirements, it is necessary to revise the matrix. However, this amendment does not work very well [24]. The BWM method does not need to compare the indicators in pairs. It is only necessary to select the best and worst indicators, and compare the best and worst indicators with the rest [23,24]. Fig. 3 illustrates the comparison principle of BWM method intuitively. The biggest advantage of BWM method is that it greatly reduces the number of comparisons. This method not only conforms to the evaluation habits of decision makers, but also greatly reduces the number of comparisons, and to a certain extent reduces the probability of comparative errors, making it more suitable for situations with relatively large standards. The specific steps of determining subjective weights with BWM can be summarized as follows:

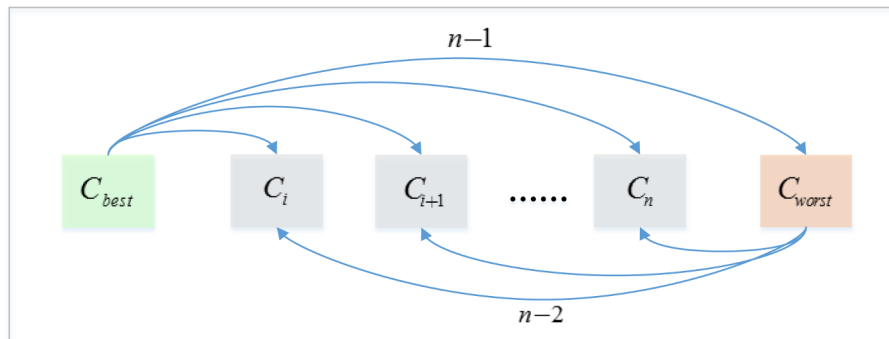


Fig. 3. The comparison principle of BWM method.

Step 1. Determine the best and worst criteria.

The best and worst are relative here. It aims to identify criteria that are more important and more valuable.

Step 2. Construct the judgment matrixs.

The scale of 1 to 9 is used to compare the relative importance of the best criterion C_{best} with other criteria and other criteria with the worst criterion C_{worst} . The judgment matrix $A_b = (a_{b1}, a_{b2}, \dots, a_{b8})$ and $B_w = (a_{1w}, a_{2w}, \dots, a_{8w})^T$ are obtained. It is obvious that $a_{bb} = 1$ and $a_{ww} = 1$.

Step 3. Calculate the optimal weight.

The optimal weight can be calculated by constructing a linear programming model. The objective function and constraints [23] are as follows:

$$\begin{aligned} & \min \xi \\ & s. t. \\ & |w_b - a_{bj}w_j| \leq \xi, \text{ for all } j \\ & |w_j - a_{jw}w_w| \leq \xi, \text{ for all } j \\ & \sum_j w_j = 1 \\ & w_j \geq 0 \end{aligned} \quad (12)$$

The optimal weight of each criterion and the value of ξ can be obtained by solving the above linear programming model. This paper would use LINGO software to solve this linear programming model.

Step 4. Conduct consistency test.

The final consistency test is carried out by calculating the consistency ratio of $\frac{\xi}{\max \xi}$ [23], where $\max \xi$ is obtained from Table 2.

4.3.2. Entropy weight method based on mixed information

The core idea of the entropy method is to analyze the difference between the index values. If the corresponding index value of a certain index is obviously different among the evaluation objects, it is considered that the index is more important in decision-making evaluation [72]. The information entropy of each criterion is calculated according to the quantitative information contained in the decision matrix. Then, according to the size of information entropy to reflect the size of the weight. The small information entropy indicates that the degree of information disorder is low, while the utility value of information is large, and the weight of corresponding indicator is large [32]. If the information entropy is large, the opposite is true. The final decision matrix in this paper contains real number, IN and TFN, so the entropy method would be developed under mixed information. The specific steps are described below.

Step 1. Calculate the average value of each evaluation criterion.

$$\tilde{f}_j = (f_{1j} \oplus \dots \oplus f_{ij} \oplus \dots \oplus f_{mj})/m = \begin{cases} \frac{1}{m} \sum_{i=1}^m f_{ij}, f_{ij} \in I_1 \\ \left[\frac{1}{m} \sum_{i=1}^m f_{ij}^L, \frac{1}{m} \sum_{i=1}^m f_{ij}^U \right], f_{ij} \in I_2 \\ \left(\frac{1}{m} \sum_{i=1}^m f_{ij}^L, \frac{1}{m} \sum_{i=1}^m f_{ij}^M, \frac{1}{m} \sum_{i=1}^m f_{ij}^U \right), f_{ij} \in I_3 \end{cases} \quad (13)$$

Table 2

The consistency test index of BWM method [24].

a_{bw}	1	2	3	4	5	6	7	8	9
$\max \xi$	0.00	0.44	1.00	1.63	2.30	3.00	3.73	4.47	5.23

where I_1 represents the attribute value of real number type, I_2 represents the attribute value of IN type, and I_3 represents the attribute value of TFN type.

Step 2. Calculate the information entropy of each criterion.

$$e_j = -\frac{1}{\ln(m)} \sum_{i=1}^m \left[\frac{d(f_{ij}, \tilde{f}_j)}{\sum_{i=1}^m d(f_{ij}, \tilde{f}_j)} \ln \left(\frac{d(f_{ij}, \tilde{f}_j)}{\sum_{i=1}^m d(f_{ij}, \tilde{f}_j)} \right) \right] \quad (14)$$

where $d(f_{ij}, \tilde{f}_j)$ represents the distance. For the specific calculation method, detail in Definition 1 and Definition 2.

Step 3. Calculate the objective weight of each criterion.

$$w_{oj} = \frac{1 - e_j}{\sum_{j=1}^n (1 - e_j)}, j = 1, 2, \dots, n. \quad (15)$$

In this paper, a simple and effective linear weighting method is used to merge subjective and objective weights. The combined weights are calculated as follows:

$$w_j = \alpha w_{sj} + \beta w_{oj} \quad (16)$$

where α and β are two parameters, satisfying $\alpha + \beta = 1$ and $\alpha, \beta \geq 0$.

4.3.3. An extended TODIM method based on mixed information

PVCS have high one-time investment costs and long payback periods, so decision makers would avoid risks and losses when choosing the suitable site. TODIM method can solve the multi-criteria decision-making problem in this kind of situation very well. The core idea of TODIM method is to consider the dominance degree of alternatives under each index, then calculate the overall dominance degree, and finally rank alternatives according to the size of dominance degree [73]. Firstly, the classical TODIM method is extended to the mixed fuzzy information environment. Secondly, different recession coefficients are set for different criteria to reflect various scenarios in real decision-making. The above extension makes the application of the TODIM method closer to reality and the analysis angle is more flexible. The specific research process will be carried out in detail through scenario analysis.

Suppose that the number of alternate sites $A = (A_1, A_2, \dots, A_m)$ is m and further evaluation criteria $C = (C_1, C_2, \dots, C_n)$ is n . The decision matrix composed of the evaluation criteria values of each alternative is $R = [r_{ij}]_{m \times n}$, and r_{ij} represents the i^{th} index evaluation value of the j^{th} alternative. r_{ij} contains three data types: real number, IN, and TFN. The steps of the TODIM method [73] based on mixed information are as follows:

Step 1. Normalize the decision matrix and calculate the relative weights of the indicators.

The standardization of quantitative indicators has been explained above, and the data of qualitative indicators has met the requirements of standardization. The normalized matrix is $F = [f_{ij}]_{m \times n}$. Subsequently, the relative weight of indicator C_j is calculated:

$$w_{jr} = w_j/w_r \quad (17)$$

$$w_r = \max\{w_j | j \in n\}$$

Step 2. Calculate the relative dominance under each criterion.

All kinds of data in decision matrix retain relatively complete evaluation information because they are not defuzzied. When calculating the gap of alternatives under a certain index, distance measure is used to reflect the difference. Therefore, the TODIM method can be well applied with using IN and TFN distance measurement methods correctly. The distance formulas of INs and TFNs have been explained

above (Eqs. (3) and (7)).

$$\Phi_j(A_i, A_k) = \begin{cases} \sqrt{\frac{w_{jr}}{\sum_{j=1}^n w_{jr}}} \tilde{d}(f_{ij}, f_{kj}) & \text{if } (f_{ij} > f_{kj}) \\ 0 & \text{if } (f_{ij} = f_{kj}) \\ -\frac{1}{\tilde{\theta}} \sqrt{\frac{\sum_{j=1}^n w_{jr}}{w_{jr}}} \tilde{d}(f_{ij}, f_{kj}) & \text{if } (f_{ij} < f_{kj}) \end{cases} \quad (18)$$

where $f_{ij} > f_{kj}$ indicates profit and $f_{ij} < f_{kj}$ indicates loss. $\tilde{d}(f_{ij}, f_{kj})$ represents the distance between two schemes and different types of values have corresponding calculation methods. $\tilde{\theta}$ is the recession coefficient. The smaller the $\tilde{\theta}$, the higher the psychological level of decision makers' loss avoidance. If $\tilde{\theta} < 1$, the impact of the loss would be increased. On the contrary, when $\tilde{\theta} > 1$, the impact of the loss would be reduced. The recession coefficient $\tilde{\theta}$ reflects the risk and loss avoidance psychology of decision makers.

Step 3. Calculate the global dominance of each alternate site.

According to the above calculation, the dominance matrix Φ_j under index C_j is constructed. The matrix form is as follows:

$$\Phi_j = [\Phi_{ik}^j] = \begin{matrix} & \begin{matrix} A_1 & A_2 & \cdots & A_m \end{matrix} \\ \begin{matrix} A_1 \\ A_2 \\ \vdots \\ A_m \end{matrix} & \begin{pmatrix} \Phi_{11}^j & \Phi_{12}^j & \cdots & \Phi_{1m}^j \\ \Phi_{21}^j & \Phi_{22}^j & \cdots & \Phi_{2m}^j \\ \vdots & \vdots & \ddots & \vdots \\ \Phi_{m1}^j & \Phi_{m2}^j & \cdots & \Phi_{mm}^j \end{pmatrix} \end{matrix}$$

where $\Phi_{ik}^j = 0$, $i, k = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$.

Then the overall dominance $\delta(A_i, A_k)$ of alternative A_i over any other alternative A_k is calculated. The formula is as follows:

$$\delta(A_i, A_k) = \sum_{j=1}^n \Phi_j(A_i, A_k) \quad (19)$$

Step 4. Sort by the comprehensive ranking value.

By standardizing the overall dominance matrix, the comprehensive ranking values ξ_i of each alternative can be obtained. The calculation formula is as follows:

$$\xi_i = \frac{\sum_{k=1}^m \delta(A_i, A_k) - \min_i \sum_{k=1}^m \delta(A_i, A_k)}{\max_i \sum_{k=1}^m \delta(A_i, A_k) - \min_i \sum_{k=1}^m \delta(A_i, A_k)} \quad (20)$$

The alternatives are sorted and preferred based on ξ_i . The larger the value of ξ_i , the better the alternative.

4.4. A location analysis and evaluation framework for urban PV charging station

As more and more PVCSSs will be laid out in cities in the future, decision makers will face the problem of selecting multiple alternative sites. Therefore, a reasonable and effective model framework can improve the efficiency of decision-making process. This section provides a location decision framework for PVCSSs as shown in Fig. 4. The decision framework consists of three main phases: Preliminary preparations, Implementation of suitability analysis and location evaluation, and Further discussion and analysis.

5. Application of the proposed model

5.1. Problem statement

The scale of promotion and application of EVs in Beijing ranks in the

forefront of China, and the level of construction of charging facilities is in the lead. In order to promote the development of EV industry and low carbon energy saving in cities, Beijing has implemented the 13th Five-Year Plan in an all-round way and vigorously promoted the construction of charging infrastructure. With the gradual enhancement of public awareness of environmental protection and the concept of green development, the PVCS will begin to be laid out in Beijing. The solar energy resources and geographical location in Beijing are shown in Fig. 5.

On the basis of market demand and policy support, an energy enterprise intends to invest in the construction of PVCS station in Beijing. In order to select a more suitable site, maximize the comprehensive utility of PVCS, the company decided to use the decision model built in this paper to optimize the site. The model integrates GIS technology, expert consultation and multi-criteria decision-making to solve this problem.

5.2. Data acquisition and processing

All kinds of data used for preliminary suitability analysis in the GIS part are shape format files. The data environment is unified into the same geographic coordinate system and then imported into ArcGIS software for operation and analysis. There are four quantitative criteria and four qualitative indicators in the multi-criteria evaluation part. Solar energy and temperature data on natural factors are obtained by reviewing the latest authoritative data. Economic indicators are obtained by calculation. Considering that these economic indicators would be affected by policy, price and rate fluctuations, this paper sets a 10% floating space to be described by IN. The criteria for judging the qualitative indicators are described in detail above, mainly through consultation with expert opinions and questionnaires. There are six experts consulted in this research, three of whom are professors from key universities in the field of energy and electricity, and the other three are professor-level senior engineers from the Research Institute of EV Development. For the collection of public acceptance evaluation information, this study obtains the evaluation information by issuing questionnaires to residents' committees in the areas where the alternative sites are located. During the one-month information collection period, the effective feedback rate of the questionnaires for each alternative site reached more than 45%. For the collection of qualitative index evaluation information, this study is based on HFLTS. The reliable sources of relevant data that need to be consulted are detailed below. The final decision matrix through information collection and processing is shown in Appendix 3.

5.3. Weight calculation for each evaluation criterion

The weight calculation of eight evaluation indexes is divided into three parts. The first part is the calculation of subjective weight, which is obtained by BWM method introduced in Section 4.3.1. In this study, LINGO software was used to solve the issue. The linear programming model constructed based on Eq. (12) is detailed in Appendix 4. The issue can be quickly solved by inputting the model into the LINGO software. The second part is the solution of objective weight. Based on the final decision matrix, the mixed information entropy method introduced in Section 4.3.2 is used to obtain the objective weight. Finally, this study sets the subjective and objective weights equally important ($\alpha = 0.5$). The combined weights of each evaluation index obtained by Eq. (16) are shown in Table 3.

It can be seen from Table 3 that the future scalability and social factors have greater weights, and the economic factors have lower weights. The results show that along with the green development and low-carbon travel concept, the layout of PVCS in Beijing is largely influenced by three factors: policy, public awareness and future development. Since the difference in solar energy resources and temperature across the whole municipality is not obvious, the importance of natural

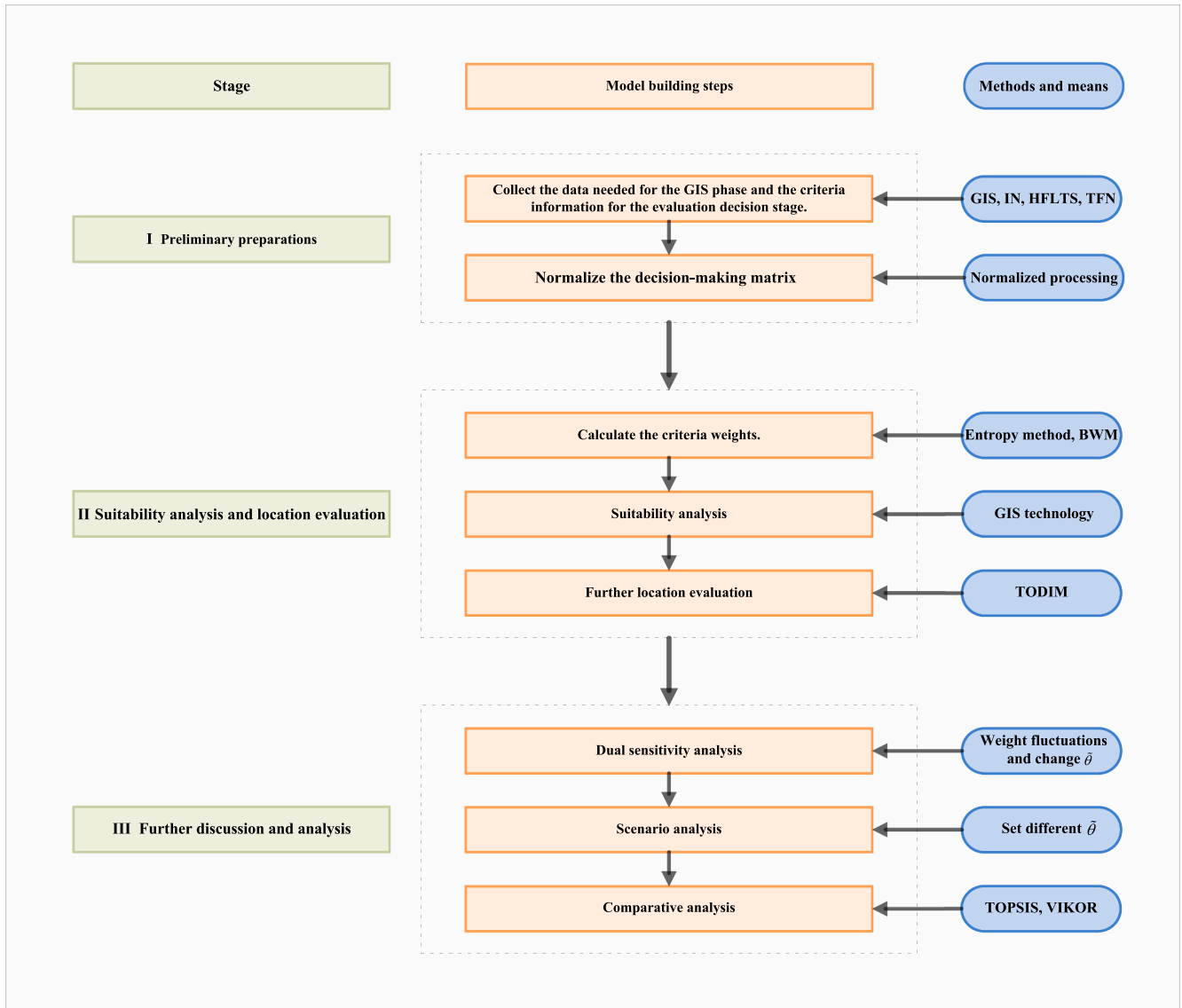


Fig. 4. The location analysis and evaluation framework for urban PV charging station.

factors is average. However, as solar energy resources and temperature are two important factors affecting PV power generation, it is also necessary to consider them when further optimizing the site. The new energy EV has been showing a rapid growth and development trend. The economy of charging station is relatively considerable under a good operation mode, so the weight of economic factors is not high. It can be considered that investors need to focus on policies, public support and potential for further development in the layout of PVCS to select suitable sites.

5.4. Suitability analysis and further location evaluation

Geographic information data are processed based on the GIS functions described in Section 4.1. As shown in Fig. 6, the suitability of available land could be quantified by calculating Euclidean Distance (ED), reclassification and suitability analysis. It should be noted that after repeated consultations by the expert group, the weights of the four types of factors in Eq. (1) are determined as: $\alpha_1 = 0.4$, $\alpha_2 = 0.25$, $\alpha_3 = 0.1$, $\alpha_4 = 0.25$. Road layout is the most important, followed by land type and gas station distribution. Because the 10-point scoring method is used in the whole process of suitability analysis, after consulting experts, this stage screens out areas with

suitability greater than 6 for further analysis: 1) considering the number of points of interest in the suitable area; 2) considering the road layout in the suitable area; 3) considering the overall layout of charging stations in the surrounding area.

After the analysis of the above steps, the seven alternative sites selected in the GIS stage are shown in Fig. 7. A1 is located near Fanqi Road in Huairou District; A2 is located near Huilongguan East Avenue in Changping District; A3 is located near Shunping Road in Tongzhou District; A4 is located near Xitucheng in Haidian District; A5 is located near Shuguang West Road in Chaoyang District; A6 is located near Guangqumen Inner Avenue in Dongcheng District; A7 is located near Xinghua Street in Daxing District.

Then, the TODIM method is used to evaluate the seven alternative sites. The key step of TODIM method is to calculate the dominance of alternatives under each index. Dominance is reflected by distance measure. The distance measure methods of interval numbers and triangular fuzzy numbers are shown in Formula Eq. (2) and Eq. (5). Here, the index C11 is taken as an example to show the dominance matrix Φ_{11} as shown in Table 4. The dominance relationship between the two schemes under the criteria C_j is calculated according to Eq. (18), wherein the recession coefficient $\tilde{\theta}$ takes a value of 1 [74–76].

Repeat the above steps and get the dominance matrix of eight

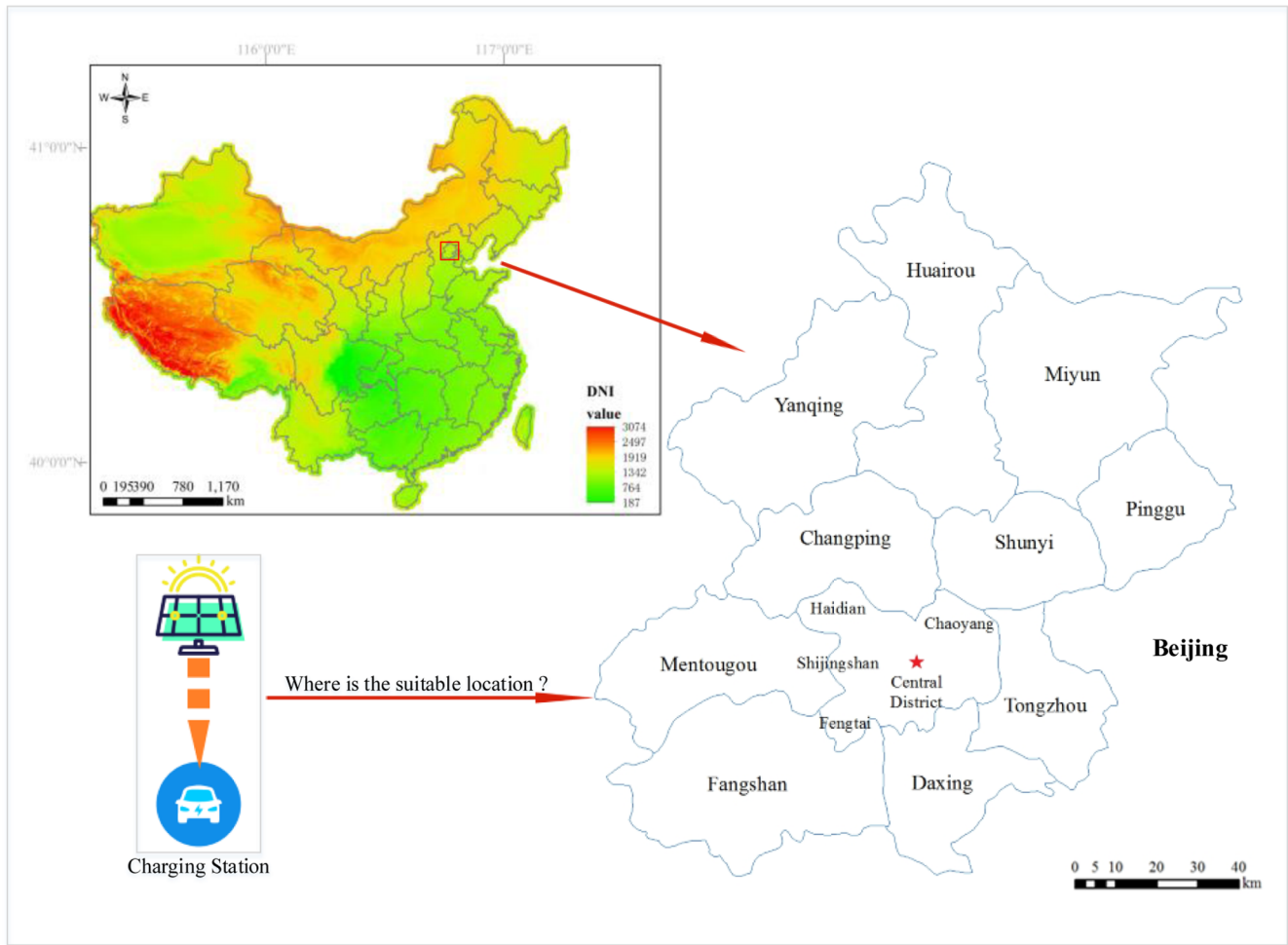


Fig. 5. Solar energy resources and geographical location in Beijing.

Table 3
The combined weight of each evaluation criteria.

Criteria	Subjective weight	Objective weight	Combination weight	Order
C11	0.1139	0.1650	0.1395	4
C12	0.0407	0.0127	0.0267	8
C21	0.0854	0.1181	0.1018	5
C22	0.0684	0.1164	0.0924	6
C31	0.1139	0.0670	0.0904	7
C32	0.2929	0.1788	0.2358	1
C41	0.1139	0.1698	0.1419	3
C42	0.1709	0.1723	0.1716	2

evaluation indicators. Then use Eq. (19) to summarize the dominance of the eight criteria to obtain the overall dominance matrix of each alternative site. According to Eq. (20), the ranking values of the alternative sites are calculated, and the results are shown in Table 5. Based on the calculation results, the order of the alternative sites can be determined as follows: $A3 > A7 > A1 > A2 > A5 > A4 > A6$. The comprehensive performance of top-ranking sites in natural, economic, technological and social aspects is excellent. Therefore, the energy planning department and investors can lay out the PVCS for Beijing based on the above analysis results. In the environment of limited resources, priority should be given to the layout of PVCS at site A3 and A7. The area covered by these sites has larger charging demand and good development conditions. Scientific and rational layout of PVCS in cities can effectively promote the use of renewable energy and reduce carbon emissions, which is consistent with the goal of energy

conversion and energy management.

The PVCS studied in this paper is suitable for the layout in such cities with good economic and social development, a certain scale market of EV and clean energy demand. The proposed model has a wide range of applications. It is not limited to the city of Beijing (China), other countries and regions can also be applied, because the location decision model constructed in this paper is universally applicable. For the Chapter 3: ①The land use type, road distribution, water distribution, gas station distribution, distribution of interest points and other indicators used in the GIS stage are all factors to be considered in the site selection of PVCS in every city. In addition, this study gives the source of geographic information data used in the GIS stage, which will be updated regularly and cover the world. ②The natural, economic, technological and social indicators constructed in the MCDM stage are also factors that need to be considered in the layout of PVCS in any region.

It is worth pointing out that the number of alternative sites is not constant. The threshold of suitability and interest points can be adjusted according to the preferences of decision makers to make the number of alternative sites more than or less than 7. The case study only takes Beijing as the research area (empirical case) to provide a complete model application process. Therefore, for the readers of this article, they can flexibly solve such problems in specific areas according to the model provided in this study.

6. Further analysis and discussion

6.1. Dual sensitivity analysis

The model built in this study mainly involves two kinds of

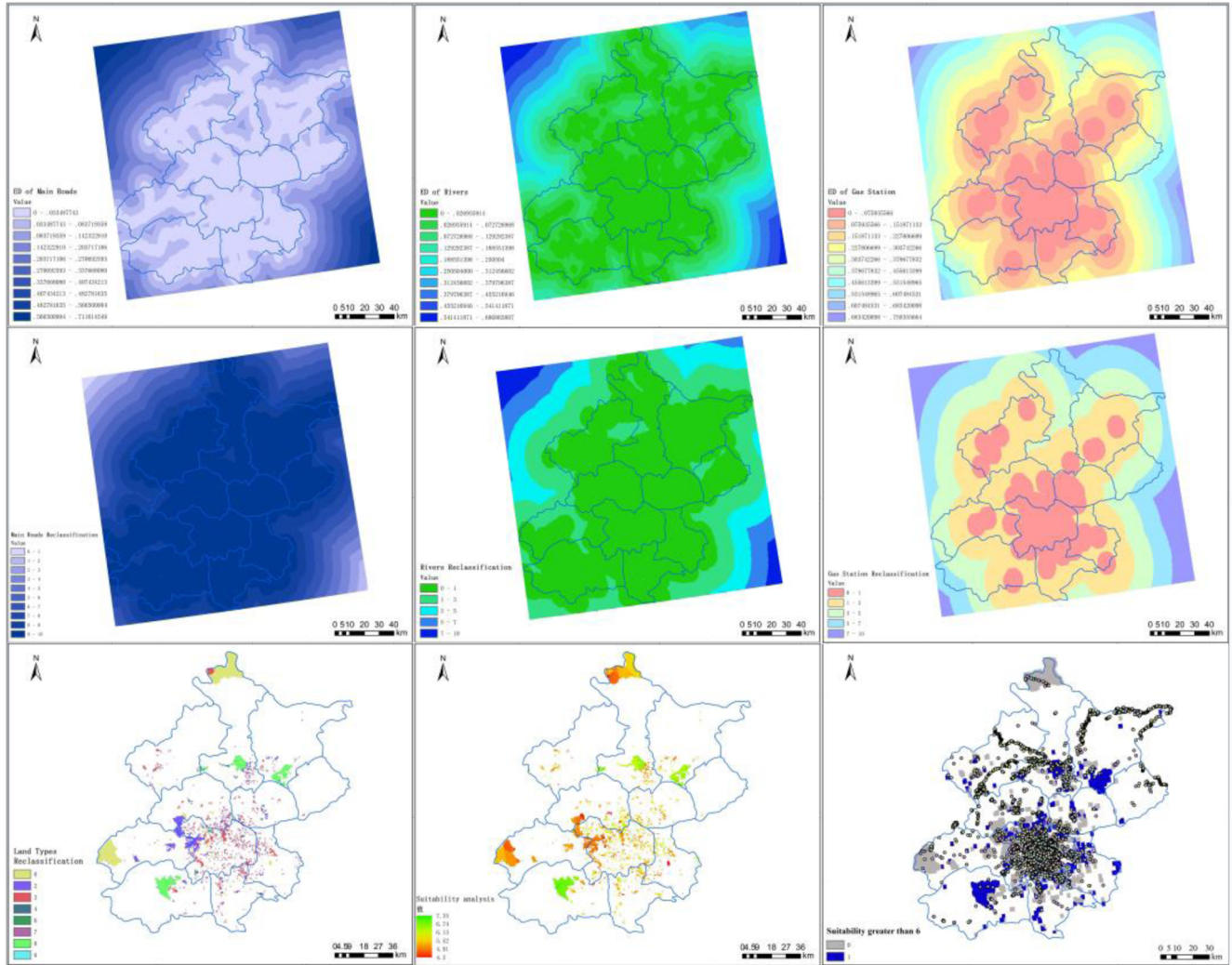


Fig. 6. Suitability analysis of the site of PV charging station in Beijing.

parameters: one is the weight of evaluation index. The multi-criteria decision-making method used in the final scheme ranking is based on the criteria weights; the other is the core parameter of TODIM method, the recession coefficient $\tilde{\theta}$. Therefore, this section would carry out dual sensitivity analysis from these two perspectives to test the stability of the model.

Part 1. Change the weights of various evaluation criteria.

In this part, a total of 56 tests were conducted to analyze the impact of the change of index weight on the results. The weights of the eight evaluation criteria will fluctuate by 10% and 20% respectively to reflect the change of weights [77]. Through this operation, the influence caused by the weight change of each index can be visually expressed, and sensitivity analysis can be effectively carried out through unified mapping.

It can be seen intuitively from Fig. 8 that the final ranking results of seven alternative sites are very stable. Overall, the curve is relatively smooth, indicating that the results are stable. The sorting results have not changed and A3 and A7 are always in the first camp, which is the most suitable area for the layout of PVCS. Separately, the alternative site A6 shows a certain sensitivity to the index C11, and the ranking score decreases slightly with the increase of weight. The reason lies in the poor performance of solar energy resources in this area compared with other alternative sites. All the schemes show good stability when the weights of index C12, C22 and C41 fluctuate. Alternate site A6 is always the area with the lowest suitability. The fluctuation range is the

most obvious in the whole test, but the range is always very small and does not exceed one unit. Therefore, it can be considered that the model built in this paper has good stability and applicability.

Part 2. Adjust the recession coefficient $\tilde{\theta}$ in TODIM method The recession coefficient $\tilde{\theta}$ is a direct parameter used in TODIM method to characterize decision makers to avoid risks and losses. The smaller the recession coefficient $\tilde{\theta}$ is, the larger the $1/\tilde{\theta}$ is; accordingly, the difference of the scheme under this index will be enlarged to show that decision makers pay more attention and sensitivity to the loss caused by this criterion. It is worth noting that such operations give all criteria the same recession coefficient. In the existing studies, $\tilde{\theta} = 1$ and $\tilde{\theta} = 2.25$ these two values are used at most [78]. In this part, the recession coefficient is set to 0.1, 0.3, 0.5, 0.7, 1, 1.3, 1.5, 1.7, 2 and 2.25[32] respectively, and then the overall dominance and ranking of schemes are calculated. The detailed data results of this sensitivity analysis are summarized in Table 6.

From Table 6, it can be seen intuitively that during the process of increasing the recession coefficient $\tilde{\theta}$ from 0.1 to 1.5, there is no change in the ranking results. However, when the recession coefficient increased from 1.5 to 1.7, the position of the top two alternative sites A3 and A7 were exchanged. This change indicates that if the sensitivity of decision makers to the differences between alternatives is very low, the ranking results would be affected. Besides, it can be seen from Fig. 9 that the smaller the recession coefficient is, the more obvious the enlargement of the gap between schemes will be, which will lead to the

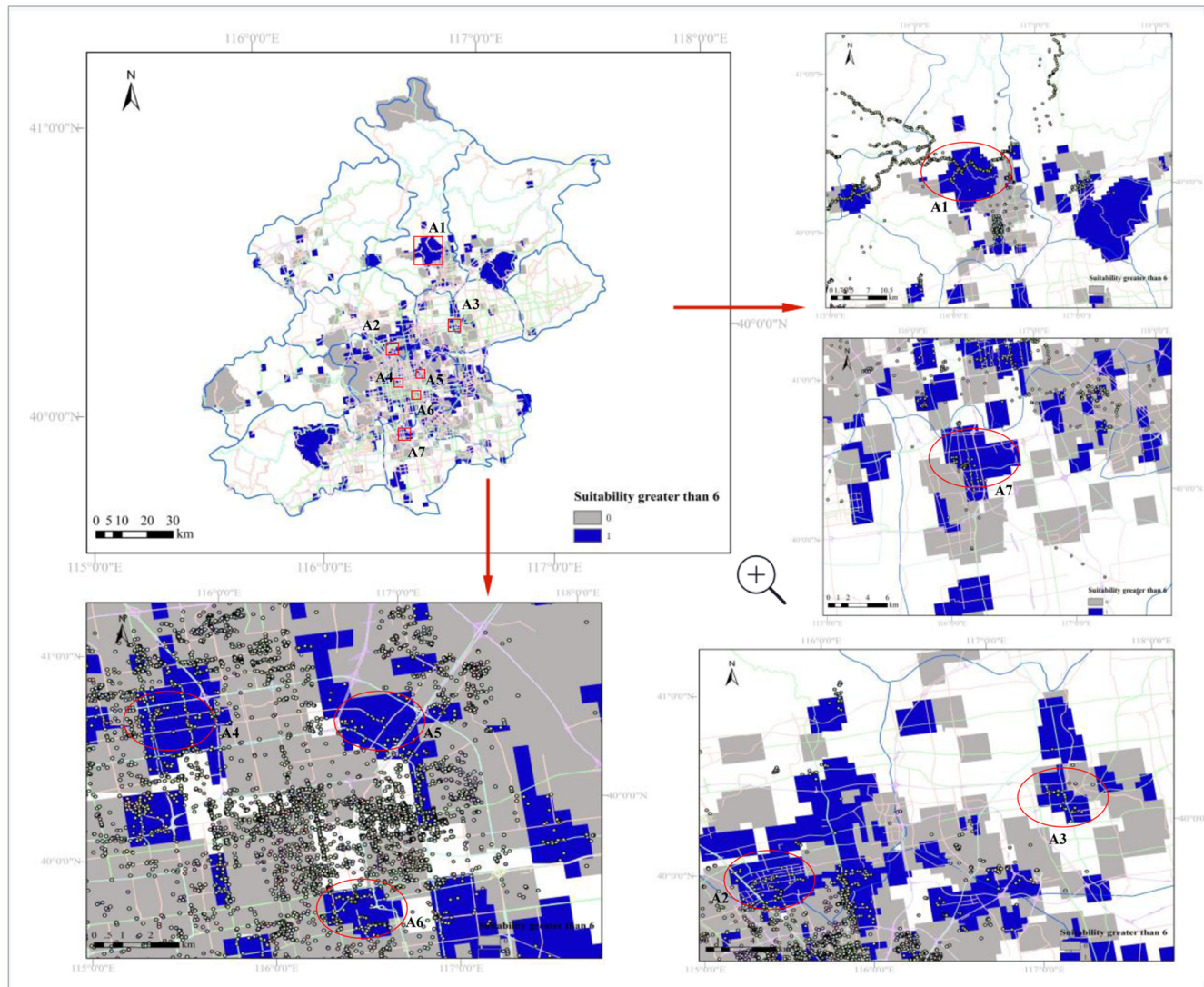


Fig. 7. Alternative sites of PVCS in Beijing.

Table 4

The dominance matrix under criteria C11.

C11	A1	A2	A3	A4	A5	A6	A7
	1.0000	0.9766	0.9948	0.9545	0.9557	0.9457	0.9448
A1	0.0000	0.0087	0.0019	0.0170	0.0165	0.0203	0.0206
A2	-0.0625	0.0000	-0.0487	0.0083	0.0078	0.0116	0.0119
A3	-0.0138	0.0068	0.0000	0.0151	0.0146	0.0184	0.0187
A4	-0.1218	-0.0593	-0.1080	0.0000	-0.0032	0.0033	0.0036
A5	-0.1186	-0.0560	-0.1048	0.0032	0.0000	0.0037	0.0041
A6	-0.1454	-0.0828	-0.1316	-0.0236	-0.0268	0.0000	0.0003
A7	-0.1478	-0.0853	-0.1340	-0.0260	-0.0292	-0.0024	0.0000

Table 5

The dominant matrix of all alternatives.

	A1	A2	A3	A4	A5	A6	A7	Summation	Standardization	Order
A1	0.0000	-0.2559	-0.9015	-0.9923	-0.9711	-0.9958	-1.0595	-5.1762	0.9195	3
A2	-1.9487	0.0000	-1.6304	-1.1787	-1.3193	-1.0121	-0.6389	-7.7282	0.6954	4
A3	-1.8640	-0.4445	0.0000	-0.4201	-0.4141	-0.1760	-0.9411	-4.2598	1.0000	1
A4	-3.4376	-1.6266	-2.4388	0.0000	-0.5281	-0.1342	-3.2109	-11.3762	0.3749	6
A5	-3.2842	-1.3341	-2.2563	-0.0763	0.0000	-0.1167	-2.2489	-9.3165	0.5559	5
A6	-4.2720	-2.6450	-3.5061	-0.9773	-1.0308	0.0000	-3.2139	-15.6450	0.0000	7
A7	-1.1434	-0.2294	-1.0293	-0.2615	-0.8651	-0.8753	0.0000	-4.4040	0.9873	2

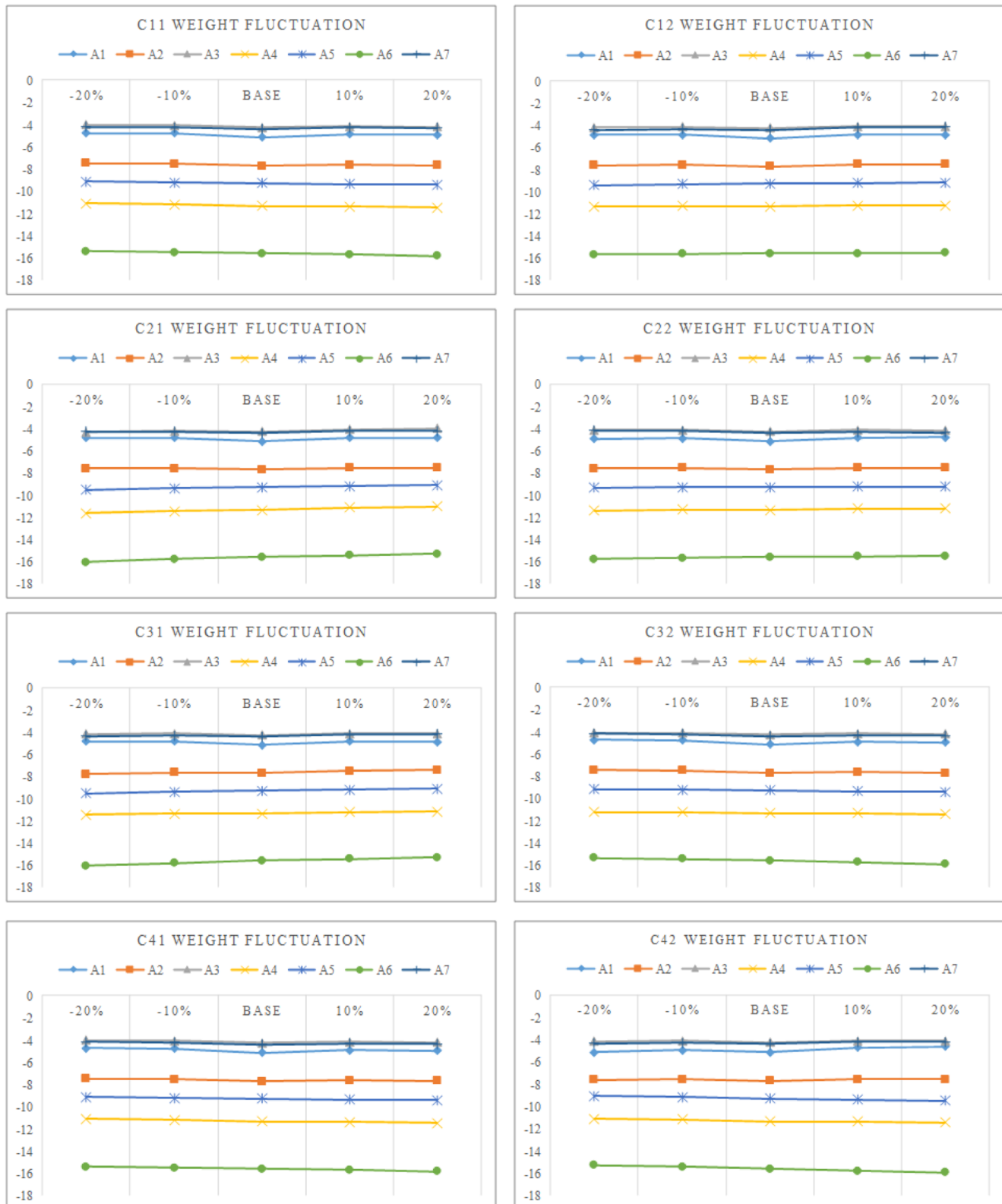


Fig. 8. Sensitivity analysis results of the Part 1.

increase of the difference of ranking values among schemes. Therefore, the above analysis shows that it is necessary to consider the decision makers' psychology of avoiding risks and losses. Generally speaking, the uniform change of the recession coefficient has no substantial impact on the final results. The model used in this paper has good stability and rationality. The scenario analysis in Section 6.2 will set different

regression coefficients for different evaluation criteria to analyze the changes in the final results.

6.2. Scenario analysis

The traditional application of TODIM is to give all the evaluation

Table 6
The schemes with different recession and order results.

	0.1		0.3		0.5		0.7		1.0	
	D	R	D	R	D	R	D	R	D	R
A1	-67.1716	3	-21.2490	3	-12.0645	3	-8.12832	3	-5.1762	3
A2	-85.7285	4	-27.9505	4	-16.3949	4	-11.4425	4	-7.7282	4
A3	-54.1758	1	-17.2010	1	-9.8060	1	-6.63674	1	-4.2598	1
A4	-120.5994	6	-39.6933	6	-23.5121	6	-16.5773	6	-11.3762	6
A5	-100.9547	5	-33.0745	5	-19.4985	5	-13.6802	5	-9.3165	5
A6	-163.5042	7	-53.9789	7	-32.0738	7	-22.6859	7	-15.6450	7
A7	-57.9389	2	-18.2834	2	-10.3523	2	-6.95326	2	-4.4040	2
	1.3		1.5		1.7		2.0		2.25	
	D	R	D	R	D	R	D	R	D	R
A1	-3.58653	3	-2.8800	3	-2.33977	3	-1.73197	3	-1.3493	3
A2	-5.72817	4	-4.8393	4	-4.15953	4	-3.39483	4	-2.9133	4
A3	-2.97989	1	-2.4110	1	-1.97605	2	-1.48667	2	-1.1785	2
A4	-8.57557	6	-7.3309	6	-6.37903	6	-5.30821	6	-4.6340	6
A5	-6.96678	5	-5.9225	5	-5.12388	5	-4.22546	5	-3.6598	5
A6	-11.8538	7	-10.1687	7	-8.88021	7	-7.43061	7	-6.5179	7
A7	-3.03128	2	-2.4212	2	-1.95466	1	-1.42981	1	-1.0993	1

Note: D and R represent dominance and ranking respectively.

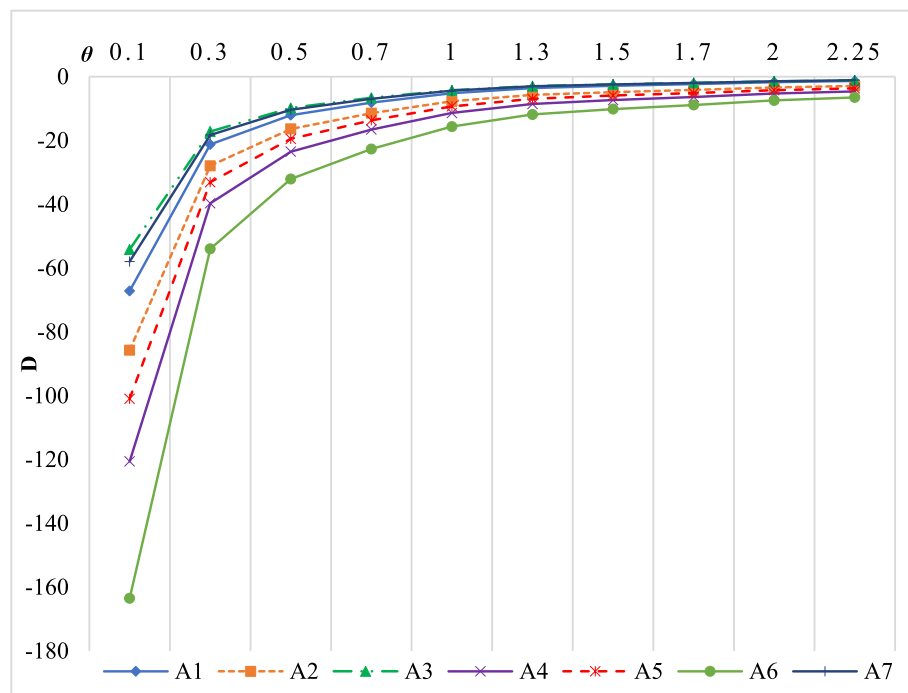


Fig. 9. Sensitivity analysis results of Part 2.

criteria the same regression coefficient (Original Settings). That is to say, decision makers have the same sensitivity and risk aversion to all kinds of evaluation indicators. However, this setting does not correspond to the actual situation. Different decision-makers have different preferences, and the same decision-maker also shows local preferences under a set of evaluation criteria. Therefore, this paper will expand the application of TODIM method and set fourteen typical scenarios to explore the changes in the final results. The regression coefficient of sensitive indicator is set to 0.1 [32]. The fourteen scenarios respectively show that the decision makers focus on the performance of one criteria group, two criteria groups, and three criteria groups. Details of the fourteen scenarios are shown in Fig. 10.

Detailed ranking values and results for fourteen scenarios are summarized as shown in Appendix 5.

It can be visually found from Fig. 11 that the ranking results in the fourteen scenarios have changed significantly compared to the original settings. In most scenarios, alternative site A6 ranks last. However, in

scenarios 4 and 7, alternative site A6 ranks relatively high. This shows that the social performance (G4) of alternative site A6 is more prominent than other performance. The ranking results under fourteen scenarios show that alternative sites A1, A7, A3 and A5 may rank first. These four sites are outstanding in all kinds of scenarios as a whole. In most scenarios, A4 and A6 rank low, so decision makers can give priority to sites A1, A3, A5 and A7 when laying out PVCs. In addition, it can be found that the model constructed in this paper can not only provide a set of final results, but also provide corresponding decision results according to different actual scenarios (preferences of decision makers). This advantage is not available in most methods.

Therefore, the following conclusions can be drawn from the above analysis: The decision maker's psychology of avoiding risk and loss (gap sensitivity) should be considered when making decision; when using TODIM method to rank schemes, it is necessary to set different recession coefficients for different indicators to reflect the decision maker's risk preference.

Criteria Group	G1	G2	G3	G4	Criteria Group	G1	G2	G3	G4
Scenario 1	0.1	1	1	1	Scenario 8	1	0.1	0.1	1
Scenario 2	1	0.1	1	1	Scenario 9	1	0.1	1	0.1
Scenario 3	1	1	0.1	1	Scenario 10	1	1	0.1	0.1
Scenario 4	1	1	1	0.1	Scenario 11	1	0.1	0.1	0.1
Scenario 5	0.1	0.1	1	1	Scenario 12	0.1	1	0.1	0.1
Scenario 6	0.1	1	0.1	1	Scenario 13	0.1	0.1	1	0.1
Scenario 7	0.1	1	1	0.1	Scenario 14	0.1	0.1	0.1	1

Fig. 10. The specific settings of scenario analysis.

6.3. Comparative analysis

The applicability and reliability of the methods used in this study need to be demonstrated by comparing with some mature and stable methods often used in existing studies. The TODIM method is the core part used in the multi-criteria evaluation model constructed in this paper. Fuzzy synthetic evaluation (FSE) is the most basic MCDM method. TOPSIS and VIKOR are mature methods often used in the field of location decision. In order to verify the reliability and applicability of this method, this section will use these three methods to calculate the problems again and compare the results.

It can be seen from Table 7 that the final results obtained by these four methods are generally consistent. A1, A7 and A3 are always the

optimal alternative sites for the layout of PVCS. Alternative A6 is always the least suitable area for layout under the three methods. The alternate sites ranked fifth and sixth are also basically the same. However, it can be found that the alternative site A3 ranks first in the TODIM method, which is different from the results obtained by the other methods. The reason is that these four methods are based on different core ideas: FSE combines index value with index weight directly; TOPSIS and VIKOR focus on the distance between positive and negative ideal schemes; TODIM pays attention to the difference between schemes under each index. The difference under each criterion would lead to the decision-maker's psychology of avoiding risk and loss. This is the focus of practical decision-making. Overall, the comparative analysis results show that the model adopted in this study has good

Scenarios								Scenarios							
1	A1	A3	A2	A7	A4	A5	A6	8	A7	A1	A3	A2	A5	A4	A6
2	A7	A1	A2	A3	A5	A4	A6	9	A7	A3	A5	A2	A1	A4	A6
3	A1	A7	A3	A2	A5	A4	A6	10	A3	A7	A5	A1	A4	A2	A6
4	A5	A4	A6	A7	A3	A2	A1	11	A7	A3	A1	A2	A5	A4	A6
5	A1	A7	A2	A3	A5	A4	A6	12	A3	A1	A7	A5	A4	A2	A6
6	A1	A3	A7	A2	A5	A4	A6	13	A3	A7	A5	A2	A1	A4	A6
7	A5	A3	A4	A6	A7	A2	A1	14	A1	A7	A3	A2	A5	A4	A6
Rank	1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	Rank	1 st	2 nd	3 rd	4 th	5 th	6 th	7 th

Fig. 11. The ranking results under fourteen scenarios.

Table 7
Ranking results by TOPSIS, VIKOR and TODIM.

	A1	A2	A3	A4	A5	A6	A7
Ranking value	0.7361	0.7000	0.7249	0.7091	0.7032	0.6760	0.7415
Ranking by FSE	2	6	3	4	5	7	1
D^+	0.0230	0.0195	0.0254	0.0383	0.0409	0.0636	0.0177
D^-	0.0539	0.0305	0.0248	0.0241	0.0178	0.0199	0.0488
Queueing value	0.7008	0.6103	0.4933	0.3860	0.3033	0.2381	0.7338
Ranking by TOPSIS	2	3	4	5	6	7	1
S_i	0.3823	0.5335	0.4027	0.5708	0.5671	0.6665	0.4713
R_i	0.1419	0.1716	0.1374	0.2358	0.1920	0.2163	0.1395
Q_i	0.0229	0.4398	0.0359	0.8316	0.6024	0.9006	0.1673
Ranking by VIKOR	1	4	2	6	5	7	3
Ranking by TODIM	3	4	1	6	5	7	2

Note: D^+ and D^- represent the distance between the scheme and the positive and negative ideal points in the TOPSIS method, respectively; S_i and R_i represent group utility and individual regret for each scheme in the VIKOR method. The smaller the Q_i , the better the corresponding scheme is.

reliability.

7. Conclusions and outlooks

With the aggravation of environmental pollution and the depletion of fossil energy, EVs are increasingly favored by the market. However, the further development of EVs still faces challenges. Firstly, the lack of reasonable layout of EVCSs in cities leads to low utilization of charging infrastructure. Secondly, EVs are directly connected to the grid for charging, which indirectly generates carbon emissions. PVCS can promote local absorption of PV power generation, while reducing indirect emissions of carbon dioxide from EVs. Based on the above situation and problems, this paper carries out the location decision research of urban PVCS.

The location decision model proposed in this paper integrates GIS and MCDM methods. The empirical analysis is carried out in Beijing as the research area. In the first stage, geographic information data such as available land type, distribution of main roads, river distribution, distribution of gas stations and distribution of interest points are used to make suitability analysis using GIS platform. In this process, the suitable location of PVCS is selected preliminarily. The second stage is multi-criteria evaluation process. In this stage, four groups of evaluation indicators are established: natural factors, economic factors, technical factors and social factors. The weight results show that social factors and future expandable potential are more important. Then, the TODIM method is used to rank seven alternative sites. The results show that priority should be given to the layout of PVCS in A3 and A7. In the part of further analysis and discussion, this paper carries out dual sensitivity analysis by adjusting the weights of indicators and the

recession coefficients of all indicators. The results show that the decision-making model constructed in this paper has good stability. In the comparative analysis part, this paper uses FSE, TOPSIS and VIKOR methods to reorder the alternative sites. The results show that TODIM method has good applicability and reliability when applied to this problem. When using the TODIM method for sorting, different regression coefficients are set for different evaluation criteria to reflect the risk preference of the decision maker. The results of scenario analysis show that considering the risk preferences in different scenarios could significantly change the final decision results. Therefore, the model provided in this paper can not only give a set of results, but also give corresponding results according to the risk preferences of decision makers. This improvement extends the usage of TODIM method.

This study provides a practical model combining decision-making methods with information technology for location decision of PVCS, and effectively improves decision-making level and work efficiency of decision-makers. Besides, it enriches the application fields of GIS and MCDM methods. Despite some contributions have been made, there are still some limitations in this paper. Based on this article, we will further deepen our research in our future work. In order to better promote the development and deployment of PVCS, we will study the capacity allocation of PV power generation and energy storage devices in PVCS, so that the entire charging station can produce optimal economic benefits. Besides, we will continue to study how to quantify the recession coefficient $\tilde{\theta}$ in TODIM, so as to clarify the exact relationship between the recession coefficient and the decision maker's psychology.

Author contributions

Jianli Zhou and Yunna Wu constructed the location decision model; Jianli Zhou and Feiyang He completed the data processing and analysis in the GIS stage. Jianli Zhou and Chenghao Wu are responsible for the multi-criteria decision-making stage. Buyuan Zhang provided modification guidance. Fangtong Liu provided software guidance. Finally, Jianli Zhou wrote the paper and formatted the manuscript for submission.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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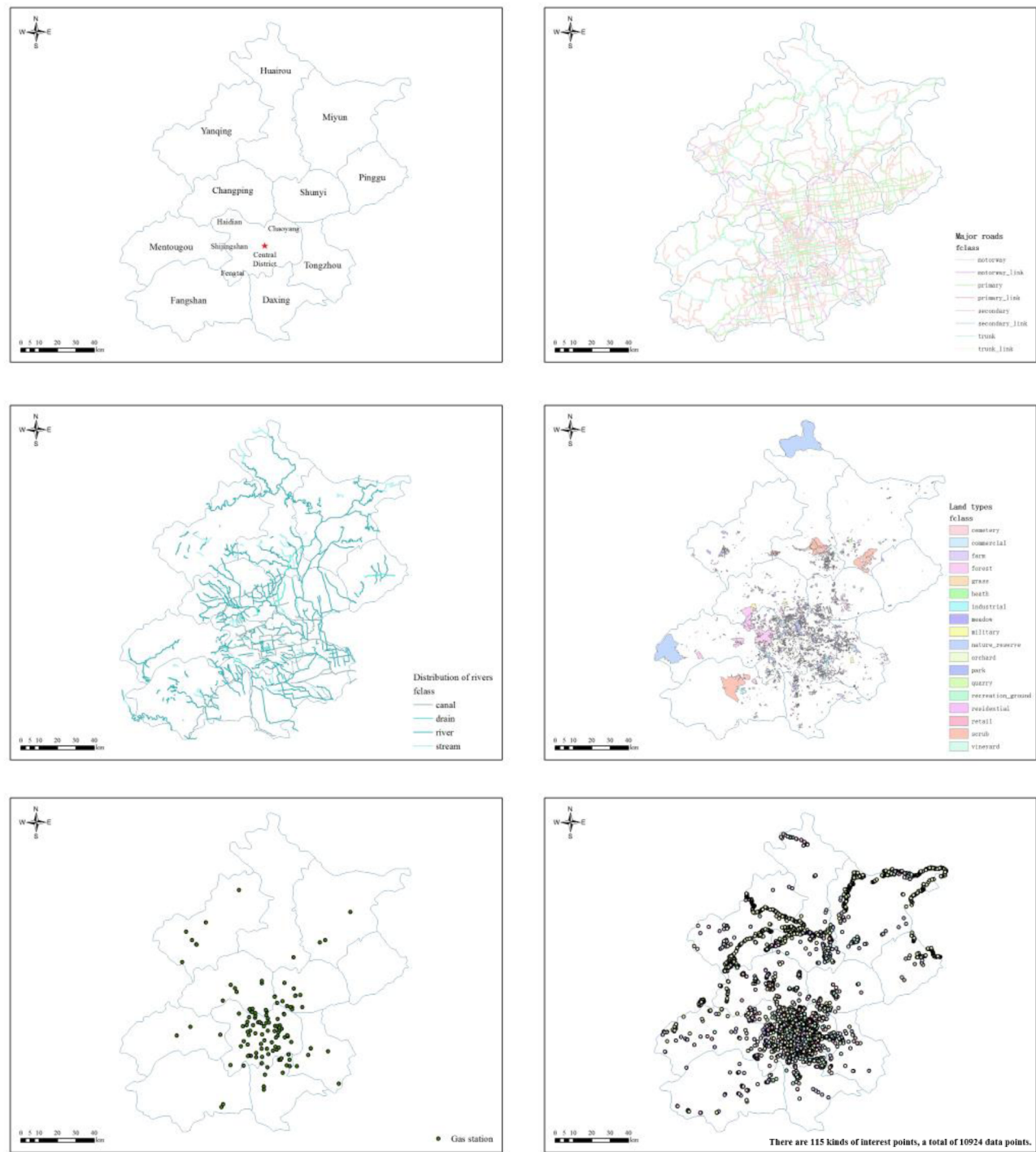
Appendix 1

The main multi-criteria decision-making (MCDM) methods used in location decision.

Category	Method	Characteristics	Research object of location decision
Weight determination	AHP	This method does not consider the correlation between indicators and the number of comparisons is large, which is easy to cause consistency problems.	Carsharing station
	ANP	This method is suitable for situations where there are many indexes and there is obvious correlation between them. The index system with less correlation would get unsatisfactory results.	Wind farm
	Entropy method	This method is the most commonly used objective weight determination method, with good stability and reliability.	Distributed PV project
	λ -measure method	This method considers the correlation between indicators, but the measurement of correlation is too rigid, which may lead to the reduction of the rationality and effectiveness of the results.	Solar thermal power plant
	DEMATEL	This method is suitable for the case of causal relationship between indicators.	Wind/solar hybrid power station

Ranking	BWM	This method greatly reduces the number of comparisons, and to a certain extent reduces the probability of comparative errors.	Biomass cogeneration project
	TOPSIS	Although this kind of method can get the final ranking value of each scheme, it cannot show the reasons why a scheme satisfies or does not meet the conditions, which is not conducive to the improvement of the scheme after ranking.	Thermal Power Plant
	VIKOR	Compared with method A, this method needs more parameters defined by decision makers, and the calculation process is more cumbersome.	Waste Disposal Plant
	ELECTRE		Offshore wind farm
	PROMETHEE	This method does not need to standardize the indicators and avoids the loss of information to a certain extent, but does not consider the psychological behavior of decision makers.	Landfill site
	TODIM	This method is a typical behavioral MCDM method. It considers that the decision-maker is limited rationality, and focuses on the decision-maker's reference dependence and loss avoidance psychology.	Distributed PV project

Appendix 2. The geographic information map generated by the relevant GIS data (©Geofabrik) used in this study



Appendix 3. Performances of each alternative on each criterion (A standardized decision matrix.)

	C1kWh/ m ²	C12°C	C21Yuan	C22Year	C31	C32	C41	C42
A1	3.3	8.9	[15058800,18405200]	[10.08,12.32]	[0.2220,0.3666,0.5336]	[0.5670,0.7498,0.9499]	[0.3333,0.5003,0.6943]	[0.4666,0.6335,0.7999]
A2	3.22	11.9	[18823500,23006500]	[9.36,11.44]	[0.3333,0.5003,0.6943]	[0.5536,0.7220,0.9332]	[0.4332,0.6002,0.7666]	[0.3780,0.5555,0.8109]
A3	3.28	11.8	[22588200,27607800]	[8.64,10.56]	[0.1943,0.4165,0.5835]	[0.4666,0.6335,0.7499]	[0.4833,0.6502,0.8165]	[0.6002,0.7666,0.9332]
A4	3.15	12.1	[30117600,36810400]	[10.8,13.2]	[0.378,0.5555,0.8109]	[0.3112,0.4722,0.7387]	[0.6502,0.8165,0.9666]	[0.6943,0.9165,1]
A5	3.15	12.3	[26352900,32209100]	[10.44,12.76]	[0.4332,0.6002,0.7666]	[0.3832,0.5501,0.7167]	[0.5833,0.8387,0.9833]	[0.6666,0.8998,1]
A6	3.12	12.3	[33882300,41411700]	[11.52,14.08]	[0.5887,0.9332,1]	[0.3499,0.5167,0.6834]	[0.6666,0.8998,1]	[0.5887,0.9332,1]
A7	3.12	12.4	[18070560,22086240]	[8.28,10.12]	[0.2832,0.4499,0.6169]	[0.6502,0.8165,0.9666]	[0.378,0.5555,0.8109]	[0.4833,0.6502,0.8165]

Natural Factor Data Sources©: Solargis.

Land Cost Data Sources©: Beijing Municipal People's Government

Appendix 4. The specific model for calculating subjective weights

$$\min = \xi$$

$$@abs(w_{C32} - 3w_{C11}) < = \xi$$

$$@abs(w_{C32} - 6w_{C12}) < = \xi$$

$$@abs(w_{C32} - 4w_{C21}) < = \xi$$

$$@abs(w_{C32} - 5w_{C22}) < = \xi$$

$$@abs(w_{C32} - 3w_{C31}) < = \xi$$

$$@abs(w_{C32} - 3w_{C41}) < = \xi$$

$$@abs(w_{C32} - 2w_{C42}) < = \xi$$

$$@abs(w_{C11} - 2w_{C12}) < = \xi$$

$$@abs(w_{C21} - 3w_{C12}) < = \xi$$

$$@abs(w_{C22} - 2w_{C12}) < = \xi$$

$$@abs(w_{C31} - 4w_{C12}) < = \xi$$

$$@abs(w_{C41} - 4w_{C12}) < = \xi$$

$$@abs(w_{C42} - 3w_{C12}) < = \xi$$

$$w_{C11} + w_{C12} + w_{C21} + w_{C22} + w_{C31} + w_{C32} + w_{C41} + w_{C42} = 1$$

$$w_{C11}, w_{C12}, w_{C21}, w_{C22}, w_{C31}, w_{C32}, w_{C41}, w_{C42} \geq 0$$

Appendix 5. Detailed ranking values and results for fourteen scenarios

	Scenario 1		Scenario 2		Scenario 3		Scenario 4		Scenario 5		Scenario 6		Scenario 7	
	D	R	D	R	D	R	D	R	D	R	D	R	D	R
A1	-5.1762	1	-15.4200	2	-6.3311	1	-55.7728	7	-15.4200	1	-6.3311	1	-55.7728	7
A2	-13.7876	3	-18.8578	3	-28.9607	4	-47.3070	6	-24.9172	3	-35.0201	4	-53.3665	6
A3	-8.5867	2	-21.7126	4	-15.4203	3	-21.2356	5	-26.0395	4	-19.7472	2	-25.5625	2
A4	-21.5683	5	-63.6139	6	-51.2807	6	-18.2650	2	-73.8060	6	-61.4729	6	-28.4571	3
A5	-22.5908	6	-48.7745	5	-46.3728	5	-11.1660	1	-62.0488	5	-59.6471	5	-24.4403	1
A6	-30.1254	7	-84.2579	7	-75.7527	7	-20.3032	3	-98.7383	7	-90.2331	7	-34.7836	4
A7	-21.3786	4	-3.8706	1	-13.6457	2	-32.2559	4	-20.8453	2	-30.6203	3	-49.2306	5
	Scenario 8		Scenario 9		Scenario 10		Scenario 11		Scenario 12		Scenario 13		Scenario 14	
	D	R	D	R	D	R	D	R	D	R	D	R	D	R
A1	-16.5750	2	-66.0166	5	-56.9277	4	-67.1716	3	-56.9277	2	-66.0166	5	-16.5750	1
A2	-40.0903	4	-58.4366	4	-68.5395	6	-79.6691	4	-74.5989	6	-64.4960	4	-46.1497	4
A3	-32.8731	3	-38.6884	2	-32.3962	1	-49.8489	2	-36.7230	1	-43.0153	1	-37.2000	3
A4	-103.5184	6	-70.5027	6	-58.1695	5	-110.4072	6	-68.3617	5	-80.6948	6	-113.7105	6
A5	-85.8308	5	-50.6240	3	-48.2223	3	-87.6803	5	-61.4966	4	-63.8984	3	-99.1051	5
A6	-144.3656	7	-88.9161	7	-80.4109	7	-149.0238	7	-94.8913	7	-103.3965	7	-158.8459	7
A7	-13.1123	1	-31.7226	1	-41.4976	2	-40.9643	1	-58.4723	3	-48.6972	2	-30.0870	2

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